AdaptiveOps — Automated E-Commerce Deployment Platform

Project Proposal

Objective

Reduce deployment complexity and improve the observability, scalability, and release reliability of a multi-service e-commerce system by implementing a repeatable, automated DevOps delivery and operations platform.

Description

Deploy Google Online Boutique, an open-source microservices store with a web frontend, a product catalog with a SearchProducts API, mock payment, and a Redis-backed cart. The team will validate the application's actual behavior and document its source attribution before deployment.

The planned workflow will process reviewed Git changes through automated container builds, Kubernetes deployment, functional checks and release verification. Monitoring, load-based scaling and rollback of a controlled faulty release will be demonstrated on Ubuntu VMs. Terraform scripts will address a feasible subset of AWS resources.

Tools & Design

- Proposed release flow: GitHub PR -> Jenkins test/build -> versioned Docker image -> Kubernetes deployment -> rollout and functional smoke checks -> accept release or roll back and verify recovery. Nginx routes traffic; Prometheus/Grafana observe it; HPA is tested separately under load.
- Infrastructure approach: local Ubuntu VMs are the primary environment; Ansible automates host configuration. Terraform will define feasible AWS resources (for example, EC2 and S3). RDS and ELB may be included if available and technically justified.

Technologies to Use

Git / GitHub; Jenkins; Docker; Kubernetes; Nginx; Prometheus; Grafana; Ansible; Terraform; AWS services; Ubuntu VM; Google Online Boutique (upstream application).

Team Members & Responsibilities

1. Amer Hossam Eldin Mohamed Mustafa (Team Leader)
Technical lead: Jenkins CI/CD, release integration, rollback; hands-on Docker, Kubernetes and monitoring.

2. Kirollos Magdy Zekry Awad
Primary focus: Kubernetes platform, Nginx routing, HPA and networking; pairs on CI/CD.

3. Rana Ashraf Abu Elyazeed Abu Elyazeed
Primary focus: Terraform feasibility, AWS resource planning, Ansible configuration; pairs on Kubernetes.

4. Rahma Mohamed Abdo El-Sharqawy
Primary focus: Prometheus/Grafana, service metrics and evidence; pairs on HPA and testing.

5. Jana Mohamed Mahmoud El-Waleely
Primary focus: Online Boutique local feasibility, Docker packaging and reproducible startup; pairs on CI.

6. Abdelrahman Ahmed Abdelbaset Mohamed
Primary focus: testing, smoke/load/rollback automation, execution notes; pairs on Jenkins/Kubernetes.

Official DEPI milestones: 16 October - first documentation; 6 November - system analysis and design; 30 November - implementation; 4 December - final testing, reports and presentation.

Week 1 (17-23 October): Baseline and feasibility

- Amer: coordinate kickoff, Git branch/PR rules, and technical acceptance checklist; pair on the local setup.
- Kirollos: check VM resource allocations and propose the simplest viable local Kubernetes approach.
- Rana: collect environment inventory; outline an Ansible host setup and AWS feasibility constraints.
- Rahma: identify useful infrastructure/application metrics and draft a small KPI checklist.
- Jana: pin an Online Boutique source revision; inspect service dependencies and attempt local startup.
- Abdelrahman: draft simple web, catalog-search API, cart, and mock-payment smoke tests.

Deliverables

- Version-pinned upstream source and a documented local startup attempt (or blockers).
- Environment inventory, initial verification checklist, and a shared Git workflow.

Week 2 (24-30 October): Containers and deployment foundation

- Amer: start a simple Jenkins test/build pipeline; pair with Jana on image versioning.
- Kirollos: prepare a local cluster and verify Pods/Services with one small deployment.
- Rana: write and locally test the first safe, repeatable Ansible configuration steps.
- Rahma: deploy a minimal Prometheus/Grafana lab or assess exporters in use.
- Jana: validate Docker builds and a documented application startup procedure.
- Abdelrahman: execute initial smoke tests and record failures for the team backlog.

Deliverables

- Repeatable application build/start instructions and preliminary Jenkins build logs.
- Minimal local Kubernetes, Ansible, and monitoring proofs of concept.

Week 3 (31 October-6 November): Analysis and design

- Amer: consolidate the architecture and CI/CD sequence.
- Kirollos: draw the Kubernetes deployment, service routing, Nginx and HPA design.
- Rana: document Ansible targets and the optional AWS/Terraform resource plan.
- Rahma: specify Prometheus metrics, Grafana dashboard layout and validation method.
- Jana: document Online Boutique services, data flow and Redis storage considerations.
- Abdelrahman: prepare use-case flows, test matrix and faulty-release recovery scenario.

Deliverables

- System Analysis & Design documentation.
- Reviewed diagrams and implementation tasks ready for the integration phase.

Week 4 (7-13 November): CI/CD and Kubernetes integration

- Amer: connect Jenkins test/build stages to a controlled Kubernetes deployment; review PRs.
- Kirollos: deploy the pinned application and configure initial Nginx entry routing.
- Rana: improve repeatability of the local Ansible host configuration.
- Rahma: collect cluster and service metrics for first operational Grafana panels.
- Jana: fix build/deployment configuration problems without redesigning the application.
- Abdelrahman: automate post-deployment smoke checks and track integration failures.

Deliverables

- Demonstrable CI-to-cluster deployment and stable Nginx access, or recorded blockers.
- Deployment smoke checks, Ansible scripts and initial monitoring screenshots.

Week 5 (14-20 November): Monitoring and scaling

- Amer: add release health gates to the pipeline and participate in dashboard review.
- Kirollos: configure Kubernetes HPA with resource requests and the required metrics source.
- Rana: validate host setup scripts; begin a low-risk Terraform proof of concept if access allows.
- Rahma: finalize Prometheus targets, Grafana views and monitoring evidence.
- Jana: investigate service/resource bottlenecks during controlled load generation.
- Abdelrahman: run repeatable load tests and document observed scale-out/scale-in behavior.

Deliverables

- Monitoring dashboards and evidence of scaling under measured load (or a tracked blocker).
- Health-check strategy and tested, repeatable configuration automation.

Week 6 (21-27 November): Recovery and full integration

- Amer: implement release failure detection and automated rollback to a known-good version.
- Kirollos: verify Kubernetes rollout history, rollback mechanics and routing after recovery.
- Rana: finish feasible Terraform/Ansible artifacts; document AWS limits and clean-up steps.
- Rahma: capture before/after incident metrics and dashboard evidence.
- Jana: support image/release reproducibility and fix critical deployment incompatibilities.
- Abdelrahman: execute faulty-deployment, rollback and recovery smoke-test scenarios.

Deliverables

- Evidence-backed rollback and recovery demo, integration test notes, and incident record.
- Terraform/Ansible files and recorded AWS feasibility/usage decision.

Week 7 (28-30 November): Stabilization and implementation submission

- Amer: lead final integration review, release tagging and source-code delivery checklist.
- Kirollos: verify final cluster, Nginx, HPA and reproducible deployment instructions.
- Rana: validate infrastructure scripts and add setup/teardown safety notes.
- Rahma: check monitoring dashboards, KPIs and exported evidence.
- Jana: verify upstream attribution, Docker build repeatability and application functionality.
- Abdelrahman: complete regression tests, bug summary and deployment walkthrough.

Deliverables

- Implementation, execution guide, scripts and source files ready.

Final integration test report, reproducible runbook